

Customer Shopping Behavior Analysis

1. Project Overview

This project analyzes the customer shopping behavior using transactional 3,900 purchases across various product categories. The goal is to uncover insights into spending purchases, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

2. Dataset Summary

- Rows: 3,900
- Columns: 18
- Key Features:
 - **Customer demographics** (Age, Gender, Location, Subscription Status).
 - **Purchase details** (Item Purchased, Category, Purchase Amount, Season, Size, Color).
 - **Shopping behavior** (Discount Applied, Promo Code Used, Previous Purchases, Frequency Of Purchases, Review Rating, Shipping Type).
 - **Missing Data**: 53 values in Review Rating Column.

3. Exploratory Data Analysis In Python

Data Preparation and cleaning in Python:

- **Data Loading**: Imported the dataset using Pandas
- **Initial Exploration**: Used `df.info()` to check structure and `.describe()` for summary statistics.

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3847.000000	3900	3900	3900	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	2	2	
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	No	No	
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	2223	2223	
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.747622	NaN	NaN	NaN	NaN	
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.715917	NaN	NaN	NaN	NaN	
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN	NaN	
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN	NaN	
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.700000	NaN	NaN	NaN	NaN	
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN	NaN	
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN	NaN	

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Customer ID                          3900 non-null   int64
1   Age                                  3900 non-null   int64
2   Gender                              3900 non-null   object
3   Item Purchased                       3900 non-null   object
4   Category                             3900 non-null   object
5   Purchase Amount (USD)                3900 non-null   int64
6   Location                             3900 non-null   object
7   Size                                 3900 non-null   object
8   Color                                3900 non-null   object
9   Season                               3900 non-null   object
10  Review Rating                        3847 non-null   float64
11  Subscription Status                  3900 non-null   object
12  Shipping Type                        3900 non-null   object
13  Discount Applied                     3900 non-null   object
14  Promo Code Used                      3900 non-null   object
15  Previous Purchases                   3900 non-null   int64
16  Payment Method                       3900 non-null   object
17  Frequency of Purchases                3900 non-null   object
dtypes: float64(1), int64(4), object(13)
memory usage: 548.6+ KB

```

- **Missing Data Handling:** Checked for null values and imputed missing values in the Review Rating column using the median rating of each product category.
- **Column Standardization:** Renamed columns to snake case for better readability and documentation.
- **Feature Engineering:**
 - Created **age_group** column by binning customer ages.
 - Created **purchase_frequency_days** column for purchase data.
- **Data Consistency Check:** Verified `discount_applied` and `promo_code_used` were redundant; dropped `promo_code_used`.
- **Database Integration:** Connected python script to PostgreSQL and loaded the cleaned DataFrame into the database for SQL analysis.

4. Data Analysis using SQL

Performed structured analysis in PostgreSQL to answer key business questions:

1. **Revenue By Gender** – Compared total revenue generated by male vs. female customers.

	gender text	Total Revenue numeric
1	Male	157890
2	Female	75191

2. **High-Spending Discount Users** – Identified customers who used discount but still spent above the average purchase amount.

	customer_id bigint	purchase_amount bigint
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62
11	24	88

3. **Top 5 Products By Rating**: Products with the Highest Average Review Ratings.

	product text	average_review_rating numeric
1	Gloves	3.85
2	Sandals	3.84
3	Boots	3.81
4	Hat	3.80
5	Jacket	3.78

4. **Shipping Type Comparison** – Compared average purchase amounts between standard and express shipping.

	Shipping Type text	average_purchase_amounts numeric
1	Express	60.48
2	Standard	58.46

5. **Subscribers vs. Non-Subscribers** – compared average spend and total revenue across subscription status.

	Subscription Status text	Average Spending numeric	Total Revenue numeric
1	No	59.87	170436
2	Yes	59.49	62645

6. **Discount-Dependent-Products** – Identified 5 products with the highest percentage of discounted purchases.

	Products text	discount_rate bigint
1	Hat	50
2	Sneakers	49
3	Coat	49
4	Sweater	48
5	Pants	47

7. **Customer Segmentation** – Classified customers into New, Returning and Loyal segments based on the purchase history.

	customer_segment text	Total Customers bigint
1	Loyal	3116
2	Returning	701
3	New	83

8. **Top 3 Products Per Category** – Listed the most purchased products within each category.

	item_rank bigint	category text	item_purchased text	total_orders bigint
1	1	Accessori...	Jewelry	171
2	2	Accessori...	Sunglasses	161
3	3	Accessori...	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

9. **Repeat Buyers & Subscriptions** – Checked whether customers with >5 purchases are more likely to subscribe.

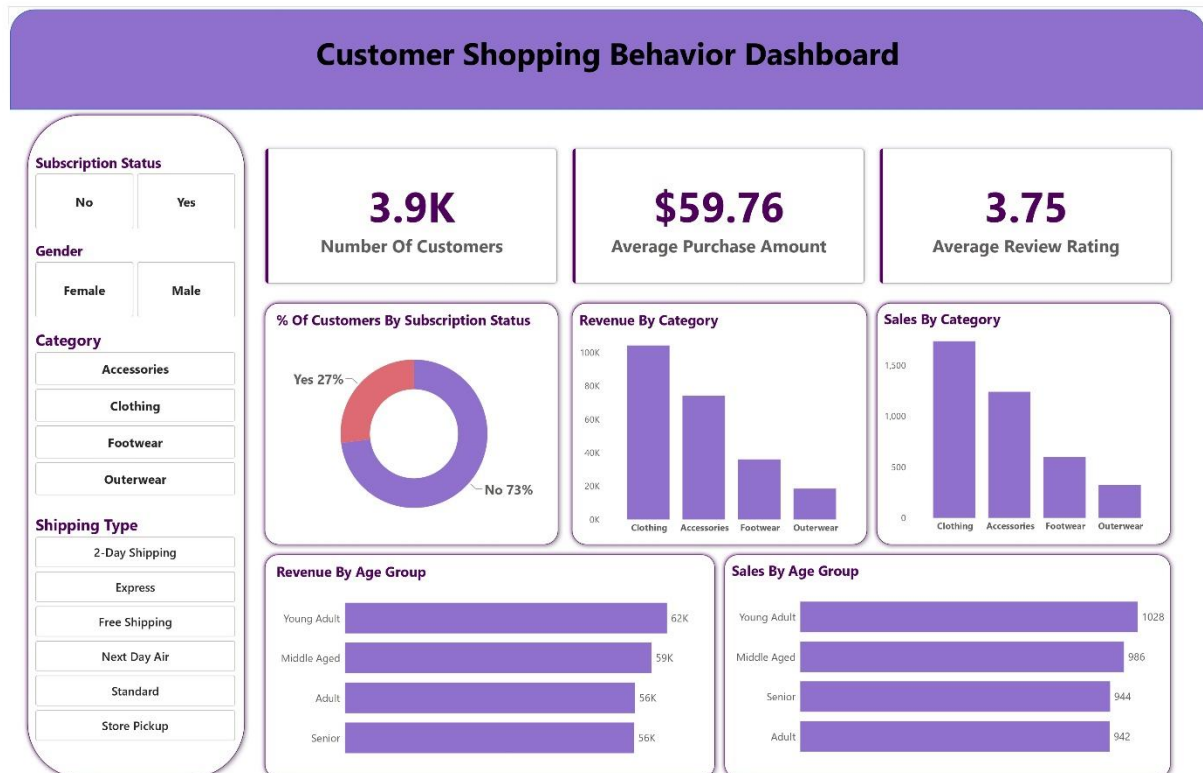
	Subscription Status text	Total Customers bigint
1	No	2518
2	Yes	958

10. **Revenue By Age Group** – Calculated total revenue contribution of each age group.

	Age Group text	Total Revenue numeric
1	Young Adult	62143
2	Middle Aged	59197
3	Adult	55978
4	Senior	55763

5. Dashboard in Power BI

Built an interactive dashboard in **Power BI** to present insights visually.



6. Business Recommendations

- **Boost Subscription** – Promote exclusive benefits for subscribers.
- **Customer Loyalty Programs** – Reward repeat buyers to move them into the 'Loyal' segment.
- **Review Discount Policy** – Balance sales boosts with margin control.
- **Product Positioning** – Highlight top-rated and best-selling products in campaigns.
- **Targeted Marketing** – Focus efforts on high-revenue age groups and express-shipping users.